「Def. Let $f:(a,b)\to\mathbb{R}$ be a map, and $x\in[a,b]$.

If the limit exists, define

$$f'(x) \overset{\text{def}}{=} \lim_{t\to x}\frac{f(t)-f(x)}{t-x} = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

This $f'(x)$ is called the derivative of f at x.

More rigorously, given $\varepsilon>0$, there exists $\delta>0$ s.t

$$0<|t-x|<\delta \Rightarrow \left|\frac{f(t)-f(x)}{t-x}-f'(x)\right|<\varepsilon \quad \text{or} \quad 0<|h|<\delta \Rightarrow \left|\frac{f(x+h)-f(x)}{h}-f'(x)\right|<\varepsilon$$

Note: Letting $h=t-x$, these are same. Sometimes, we can use the following equivalent:

「Lemma. Let $f:(a,b)\to\mathbb{R}$ be a map, and $x\in[a,b]$. TFAE:

a) $f'(x)$ exists.

b) There exists a map $\eta: B_\delta(0)\setminus\{0\}\to\mathbb{R}$ for some $\delta>0$ and $L\in\mathbb{R}$ s.t

$$f(x+h)-f(x)=L\cdot h+|h|\cdot\eta(h) \quad \text{and} \quad \eta(h)\to 0 \text{ as } h\to 0.$$

Proof. a)⇒b) Given $\varepsilon>0$, there exists $\delta>0$ s.t

$$0<|h|<\delta \Rightarrow \left|\frac{f(x+h)-f(x)}{h}-f'(x)\right|<\varepsilon.$$

Now, let $L=f'(x)$ and define for each $0<|h|<\delta$

$$\eta(h)=\frac{1}{|h|}\cdot\left(f(x+h)-f(x)-L\cdot h\right)$$

Then $\lim_{h\to 0}\eta(h)=0$. Conversely, if such η exists, then

$$\frac{f(x+h)-f(x)}{h}=L+\frac{|h|}{h}\cdot\eta(h)\to L \text{ as } h\to 0,$$

therefore a) is satisfied

Note: the absolute value above η is Not necessary in this case, but it is necessary when our ambient space is More $\mathbb{R}^1$, is $\mathbb{R}^p$.

Thm. Let $f, g : (a,b) \to \mathbb{R}$ be maps, and let $x \in (a,b)$.
If f, g are differentiable at x, then $f+g$, $f-g$, f/g are differentiable at x with

1) $(f+g)'(x) = f'(x) + g'(x)$
2) $(f \cdot g)'(x) = f'(x)g(x) + f(x)g'(x)$
3) $(f/g)'(x) = \left(f'(x)g(x) - f(x)g'(x)\right) / g(x)^2$

Proof. Given $\varepsilon > 0$, there exists $\delta > 0$ s.t

$$\left| \frac{f(x+h) - f(x)}{h} - f'(x) \right|, \left| \frac{g(x+h) - g(x)}{h} - g'(x) \right| < \frac{\varepsilon}{2} \quad \text{if } 0 < |h| < \delta.$$

This implies

$$\left| \frac{f(x+h) + g(x+h) - (f(x) + g(x))}{h} - (f'(x) + g'(x)) \right| < \varepsilon$$

By the triangle inequality. To show 2), observe that

$$f(x+h)g(x+h) - f(x)g(x) = f(x+h) \cdot (g(x+h) - g(x)) + g(x) \cdot (f(x+h) - f(x))$$

Dividing both side by h, and letting $h \to 0$. Then we obtain the result.
Similarly, observe that

$$\frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)} = \frac{f(x+h)g(x) - f(x)g(x+h)}{g(x+h) \cdot g(x)}$$

$$= \frac{1}{g(x+h) \cdot g(x)} \cdot \left[(f(x+h) - f(x)) \cdot g(x) - f(x)(g(x+h) - g(x)) \right]$$

Dividing... Yeah!

Thm. Let $f: (a,b) \to \mathbb{R}$ and $g: J \to \mathbb{R}$ be functions, where $\operatorname{Im} f \subseteq J$, and let $x \in (a,b)$. If $f'(x)$ exists and $g'(f(x))$ exists, then $g \circ f$ is differentiable at x with

$$(g \circ f)'(x) = g'(f(x)) \cdot f'(x)$$

Proof. Using the equivalent condition for differentiable: there exists η_1, η_2 s.t

$$f(x+h_1) - f(x) = h_1 \cdot f'(x) + |h_1| \cdot \eta_1(h_1) \quad \text{and} \lim_{h_1 \to 0} \eta_1(h_1) = 0$$

$$g(f(x)+h_2) - g(f(x)) = h_2 \cdot g'(f(x)) + |h_2| \cdot \eta_2(h_2) \quad \text{and} \lim_{h_2 \to 0} \eta_2(h_2) = 0.$$

Now, let $h_2 = f(x+h_1) - f(x)$ then (It is possible since $f(x+h_1) - f(x) \to 0$ as $h_1 \to 0$

$$(g \circ f)(x+h_1) - (g \circ f)(x) = g(f(x)+h_2) - g(f(x)) = h_2 g'(f(x)) + |h_2| \cdot \eta_2(h_2)$$

Dividing by h_1 and letting $h_1 \to 0$ then

$$\frac{(g \circ f)(x+h_1) - (g \circ f)(x)}{h_1} = \frac{h_2}{h_1} \cdot g'(f(x)) + \frac{|h_2|}{h_1} \cdot \eta_2(h_2) \to f'(x) \cdot g'(f(x))$$

Problem. Let $f:\mathbb{R}\to\mathbb{R}$ be a differentiable with $\sup|f'(x)|<1$.
Then f has a fixed point.

Proof. Construct a sequence. Let $x_1=\alpha\in\mathbb{R}$, $x_{n+1}=f(x_n)$ $n\geq 1$.
Take $\beta=\sup|f'(x)|<1$. Given $n\in\mathbb{N}$, if $x_n=x_{n-1}$ then x_{n+1} is the fixed point. If $x_n=x_{n-1}$,

$$|x_{n+1}-x_n|=|f(x_n)-f(x_{n-1})|=|x_n-x_{n-1}|\cdot|f'(c)|\leq\beta\cdot|x_n-x_{n-1}| \quad \text{by the MVT.}$$

Now, $|x_{n+1}-x_n|=|f(x_n)-f(x_{n-1})|\leq\beta|x_n-x_{n-1}|$
$=\beta\cdot|f(x_{n-1})-f(x_{n-2})|\leq\beta^2|x_{n-1}-x_{n-2}|$
$\vdots$
$\leq\beta^{n-1}|x_2-x_1|$.

Given $\varepsilon>0$, put $N\in\mathbb{N}$ such that $\beta^{N-1}|x_2-x_1|<\varepsilon(1-\beta)$
Then for any $m\geq n\geq N$,

$$\begin{aligned}|x_m-x_n|&\leq|x_m-x_{m-1}|+|x_{m-1}+x_{m-2}|+\cdots+|x_{n+1}-x_n|\\&\leq\left(\beta^{m-2}+\beta^{m-1}+\cdots+\beta^{n-1}\right)\cdot|x_2-x_1|\\&=\beta^{n-1}\left(1+\beta+\cdots+\beta^{m-n-1}\right)\cdot|x_2-x_1|\\&<\beta^{n-1}\left(\sum_{k=0}^{\infty}\beta^n\right)\cdot|x_2-x_1|=\beta^{n-1}\cdot\frac{1}{1-\beta}\cdot|x_2-x_1|<\varepsilon.\end{aligned}$$

Therefore $\{x_n\}$ is cauchy sequence in $\mathbb{R}$, the complete metric space, thus

$$x^*=\lim_{n\to\infty}x_n$$

exists. Finally, the differentiable map continuous,

$$f(x^*)=f\left(\lim_{n\to\infty}x_n\right)=\lim_{n\to\infty}f(x_n)=\lim_{n\to\infty}x_{n-1}=x^*.$$

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